■■ EmotiBeat: Complete Spoken Interview Script

A Step-by-Step Word-for-Word Speaking Guide to Explain Your Project to Interviewers

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PART 1: The Opening & Hook (When they ask: 'Tell me about your project')

■ **INTERVIEW ACTION / CUE: Maintain eye contact, speak with energy and clarity.**

YOU SAY:

"Good morning / afternoon! I'd love to explain my project, **EmotiBeat**.

EmotiBeat is an **AI-powered, real-time emotion-controlled music player**. The core idea is to eliminate manual search and playlist fatigue by allowing the system to dynamically sense the user's emotional state in real time and automatically stream music that matches or elevates their mood.

What makes this project unique is that it doesn't just rely on a simple face detector; it combines **Computer Vision, Vision Transformers (ViT), ONNX Deep Neural Networks, Google Gemini GenAI for multilingual text sentiment, and a Temporal Smoothing Engine** to create a seamless, jitter-free music listening experience."

■ *Interview Pro-Tip: Notice how this opening highlights AI, GenAI, signal processing, and audio in under 45 seconds.*

PART 2: Why I Built This & The Problem Statement

■ **INTERVIEW ACTION / CUE: Show empathy for user experience and explain the technical gap in existing apps.**

YOU SAY:

"If we look at popular music apps like Spotify or Apple Music, recommendations are largely based on past listening history or static genre tags. But human emotions are dynamic — you might be stressed during work, celebrating good news, or feeling exhausted in the evening.

Existing research projects in emotion-based music often fail because of two big flaws:
1. Emotion flickering: A single blink or lighting shift causes the music to stop and switch songs abruptly.
2. Strict single-modality: If the user turns off their webcam or is in a dark room, the entire app becomes useless.

I engineered EmotiBeat to solve both of these real-world challenges through **dual-modal inputs and temporal stabilization**."

■ *Interview Pro-Tip: Interviewers love candidates who identify why standard naive implementations fail.*

PART 3: End-to-End Technical Pipeline (Step-by-Step)

■ **INTERVIEW ACTION / CUE: Break the architecture into 5 logical chronological steps.**

YOU SAY:

"Let me walk you through the end-to-end data flow from the moment a user appears on camera to when the song plays:

Step 1: Face Detection & Intelligent Margin Expansion

The camera captures the frame. Using OpenCV, I perform histogram equalization to handle dim or bright lighting. Then, instead of taking a tight bounding box, I engineered a **+15% margin expansion**. This is crucial because emotions like sadness or surprise rely heavily on eyebrow raises, forehead wrinkles, and jawline movement which tight crops cut off.

Step 2: Dual Deep Learning Inference (ViT & ONNX)

The cropped face is passed into our classification engine. I provided two switchable backends: a **Vision Transformer (ViT)** for state-of-the-art attention-based expression accuracy, and an **OpenCV FERPlus ONNX model** optimized for sub-15ms inference on ordinary CPUs. It classifies 5 primary emotions: Happy, Sad, Angry, Surprise, and Neutral.

Step 3: Multimodal GenAI & Multilingual Text Mode (Google Gemini)

If a user prefers typing how they feel or their webcam is off, they can write their thoughts in English, Hindi, or Hinglish (e.g., *'Bohot stressed hu aaj exam ki wajah se'*). I integrated Google Gemini (ChatGoogleGenerativeAI) with zero temperature to perform semantic sentiment reasoning and extract the exact emotional mood and song theme.

Step 4: Temporal Smoothing & Hysteresis Filtering

This is the brain of the player. To prevent songs from constantly jumping, raw probabilities pass through an **Exponential Moving Average (EMA)** with a decay factor of 0.65, followed by an **N=10 Sliding Window Consensus Buffer**. A song only switches if the dominant emotion maintains a stability score greater than 70%.

Step 5: Dynamic Music Recommendation & Procedural Synthesis

Once the mood is confirmed, our Extensible Music Provider queues the matching song. We support **39 curated multilingual YouTube tracks**, local mp3/wav libraries, and a **built-in Procedural Harmonic Synthesizer** that algorithmically generates audio waveforms in memory so the project works out-of-the-box even without an internet connection."

■ *Interview Pro-Tip: Mentioning ViT, ONNX, EMA, and Gemini shows you understand both edge computing and modern GenAI.*

PART 4: Key Technical Challenges You Solved

■ **INTERVIEW ACTION / CUE: Pick this story to demonstrate deep debugging and analytical problem-solving.**

YOU SAY:

"One of the most interesting bugs I encountered during testing was the **'Teeth Grimace vs. Smile' false positive**.

When people are angry, frustrated, or crying in pain, they often clench or bare their teeth. Because standard FER training datasets correlate visible teeth with smiling, standard neural networks would classify intense distress as 'Happy' with 40-50% confidence!

To fix this, I engineered a custom **Distress Filter heuristic** in src/model.py: if secondary distress signals (fear, disgust, sadness) are elevated above 0.25, the happy probability is mathematically penalized unless the smile-to-distress ratio is decisively greater than 1.4. This immediately eliminated false happy triggers during genuine frustration."

■ *Interview Pro-Tip: Interviewers love hearing about real heuristics that fix domain-specific edge cases.*

PART 5: What to Say While Giving a Live Demo

■ **INTERVIEW ACTION / CUE: Narrate your screen while running 'streamlit run app.py'.**

YOU SAY:

"*(Opening the browser)*: 'Here is the live EmotiBeat dashboard, built with a modern dark glassmorphism theme in Streamlit.

On the left column, you can see our live camera feed with real-time face bounding boxes and confidence meters. Notice that as I smile, the confidence meter smoothly climbs to Happy (95%), and the player queues an upbeat track.

If I briefly look away or blink, watch the stability index — the Temporal Smoother prevents the song from resetting because it knows it's a micro-expression.

Now let's switch over to the **Text & Gemini GenAI Tab**. If I type: *'Feeling exhausted after studying all night'*, Gemini classifies this as Sad/Melancholic and automatically triggers relaxing lofi acoustic music. All components are completely tested with our automated Pytest test suite.'"

■ *Interview Pro-Tip: Always talk while clicking; never let there be awkward silence during a demo.*

PART 6: Strong Final Closing Statement

■ **INTERVIEW ACTION / CUE: Conclude with confidence and invite architectural discussion.**

YOU SAY:

"To summarize: EmotiBeat is not just a toy machine learning demo. It is a full-stack, modular audio intelligence system that brings together **Vision Transformers, ONNX edge inference, Multimodal GenAI, temporal signal stabilization, and extensible audio streaming**.

The code is fully modular, adhering to clean OOP principles, and 100% of unit tests pass cleanly. I would be happy to dive into any specific module, loss functions, or architecture decisions you'd like to explore!""

■ *Interview Pro-Tip: This invites the interviewer to ask questions in areas where you are strongest.*